

Ukemi-SafeFall: An Ukemi-inspired Injury-Aware Falling and Recovery

Abstract—Humanoid robots remain vulnerable to falls when balance recovery fails. Controllers that treat falling as terminal failure leave an open safety problem: how to reduce mechanical damage during unavoidable falls while preserving the option to stand up afterward. This paper proposes Ukemi-SafeFall, an ukemi-inspired, injury-aware falling and recovery framework that shapes humanoid fall-recovery behavior through condition-gated rolling and impact-mitigation rewards. In simulation, Ukemi-SafeFall reduces injury-risk surrogates of normalized peak contact force, cumulative peak contact force, and unsafe-contact events by 32%, 78%, and 70%, respectively, compared with passive falling. Experiments further demonstrate that Ukemi-SafeFall transfers to a humanoid platform across a range of falling scenarios. Supplementary videos and project materials will be made available at: <https://ukemi-safefall.github.io/>

I. INTRODUCTION

Humanoid robots are increasingly expected to operate in dynamic, human-centered environments where they must walk, run, manipulate objects, and respond to unexpected disturbances. Although advances in balance and push-recovery control have improved robustness, unexpected disturbances, modeling errors, perception failures, and hardware faults can still lead to an irrecoverable loss of balance [1], [2]. Because uncontrolled impacts can produce high contact forces, vulnerable head or torso collisions, and large joint loads that threaten hardware [3], [4], fall management must include not only prevention but also controlled falling and post-fall recovery [1].

To address the critical safety challenge of unavoidable falls, extensive research has explored both fall prediction mechanisms [1]–[3], [5]–[7] and controlled impact minimization [4], [8]–[15]. However, despite these advancements in active impact mitigation, existing methods still tend to treat the fall or the recovery as an isolated final objective. Consequently, there remains a lack of end-to-end frameworks that seamlessly bridge safe ground-contact management and autonomous post-fall recovery.

To bridge this gap, our work draws inspiration from *ukemi* [16], martial-arts breakfall techniques designed to reduce hazardous loading during a fall while maintaining the posture and momentum necessary for immediate recovery. Biomechanical studies [17]–[19] associate these techniques with changes in head acceleration and impact distribution (Fig. 1). Building on these concepts, this paper addresses continuous humanoid fall management by introducing measurable *injury-risk surrogates*—namely normalized peak contact force, cumulative peak contact force, and unsafe contact events. By explicitly quantifying and minimizing these metrics, our approach actively mitigates



Fig. 1. A humanoid robot (bottom) performs ukemi-inspired falling and recovery maneuvers for unavoidable falls, inspired by ukemi techniques demonstrated by a human [16] (top).

simulated injury during impact and conditions the robot’s state for a seamless transition into post-fall recovery.

To this end, we propose **Ukemi-SafeFall**, an ukemi-inspired, injury-aware falling and recovery framework that shapes humanoid fall-recovery behavior through condition-gated rolling and impact-mitigation rewards. These rewards encourage proactive roll entry, soft landings, and strategic contact redistribution to protect fragile extremities and joints, while recovery-oriented rewards ensure the robot actively seeks an upright, stable posture after the fall ends.

To ensure these highly dynamic maneuvers remain natural and kinematically fluid, we regularize the policy using a Score-Matching Motion Prior (SMP) [20]. Leveraging this prior, we further use Generative State Initialization (GSI) [20] to seed training episodes directly from diverse fallen configurations, thereby accelerating the agent’s exploration of post-fall recovery behaviors. Crucially, while the SMP multiplicatively gates the overall reward to enforce realistic motion, the specific impact-mitigation strategies and protective *ukemi* behaviors are driven by our condition-gated reward terms. The contributions of this work are summarized as follows:

- We propose a training framework that bridges the gap between falling impact mitigation and post-fall recovery. By integrating condition-gated rolling and impact-mitigation rewards, the system continuously manages a fall from initial impact through a stable standing state.
- We introduce quantitative injury-risk surrogate metrics and a fallen-state evaluation protocol that measures both injury-risk and recovery surrogates.
- We validate the framework in simulation and demon-

strate successful transfer to a humanoid robot across a range of falling scenarios.

II. RELATED WORK

Research related to Ukemi-SafeFall spans fall prediction and protective strategies, controlled falling and impact minimization, learning-based post-fall recovery, and data-driven motion priors. As inspiration for this work, *ukemi* breakfall techniques [17]–[19] aim to redirect and distribute impacts through rolling and controlled multi-contact, and they have been studied in human biomechanics as an injury-mitigation strategy.

A. Fall Prediction and Protective Strategies

Fall-prediction methods estimate whether balance recovery remains feasible from recent state histories, external disturbances, or fault-induced deviations [1], [2], [5], [6]. Protective-policy architectures [7] use these estimates to switch from a nominal task policy to a dedicated mitigation controller. SafeFall [3] similarly combines fall prediction with protective control to reduce contact forces, joint loads, and contact with vulnerable robot components. Our work does not learn a fall predictor or a policy-switching mechanism; instead, it studies an active fall-management controller, focusing on ukemi-inspired rewards that guide both the landing and recovery phases.

B. Controlled Falling and Impact Minimization

Classical controlled-falling methods [4], [8] regulate body attitude so that contact occurs through selected shock-absorbing or structurally robust regions. Other approaches optimize the falling trajectory [9], [10], alter fall direction through stepping and inertia shaping [11], or plan multiple contacts to reduce impact damage [12], [13]. Related methods [14] design protective postures and contact sequences to reduce estimated fall damage. Learning-based safe-falling methods later formulated policies that choose contact sequence, location, and timing to reduce peak impact measures [15]. These studies establish important mechanisms for body reorientation, contact planning, and impact mitigation. Our work builds on these mechanisms by integrating condition-gated rolling for reorientation, soft-landing strategies for contact planning, and explicit force objectives for impact mitigation, all within an unified policy that also performs post-fall recovery.

C. Learning-Based Fall Mitigation and Recovery

Recent learning-based systems extend fall management beyond trajectory-level impact optimization. SafeFall [3] learns a protective response for full-scale humanoids but does not target post-fall recovery. Unified Humanoid Fall-Safety [21] learns prevention, impact mitigation, and recovery behaviors in an unified policy from sparse demonstrations. Dedicated get-up controllers [22], [23] recover from diverse fallen poses and terrain conditions, but they typically do not optimize the preceding falling motion and resulting impact. Unified Humanoid Fall-Safety is the most comparable prior work in

terms of overall scope. Our approach, however, is distinct due to its explicitly condition-gated, ukemi-inspired rolling and landing objectives, the use of injury-risk surrogate metrics, and the incorporation of a frozen motion prior. Because our method differs from theirs in motion data, initialization, disturbance setup, and evaluation protocol, we do not regard the results as directly comparable.

D. Motion Priors for Humanoid Control

In humanoid control, motion priors act as regularizers for learned policies, ensuring that generated behaviors remain natural, expressive, and physically plausible. Reinforcement learning (RL)-based motion-tracking frameworks such as DeepMimic [24] and BeyondMimic [25] learn robust physics-based skills by tracking reference motions. However, strict frame-by-frame tracking objectives can limit generalization to new tasks. Distribution-matching techniques offer greater adaptability. For instance, Adversarial Motion Priors (AMP) [26] extract behaviors from reference motions via adversarial style matching. However, AMP remains computationally demanding, as it requires persistent access to the full dataset and joint training of a discriminator for each new policy. Score-Matching Motion Priors (SMP) [20] instead pre-train a reusable, frozen diffusion prior and use score distillation as a task-agnostic motion reward, leveraging modern diffusion and score-based generative modeling [27], [28]. Alongside such reward-based priors, advanced state-seeding schemes such as Reference State Initialization (RSI) [24] and Generative State Initialization (GSI) [29] are widely adopted to improve training efficiency and accelerate the acquisition of complex behaviors. In our work, we combine SMP with GSI to develop a robust and computationally efficient fall-mitigation policy that generalizes across a broad range of impact conditions.

III. HARDWARE PLATFORM

All hardware experiments are conducted on a humanoid robot measuring 1.11 m in height and weighing 32 kg. The robot has 23 actuated DoFs, distributed as 12 in the legs, 10 in the arms, and 1 in the waist yaw. The platform is designed for durability and low manufacturing cost, making it well suited for repeated fall experiments. Because Ukemi-SafeFall induces highly dynamic full-body rolling, unmodeled internal risks (e.g., cable strain, gearbox shock, and component impacts within the shell) may still occur even when external contact forces are reduced; to mitigate this, we enforce conservative actuator torque limits, include torque/velocity shaping terms in the reward, and use protective covers on protruding components during experiments. Actuator torque and speed limits, link inertias, and segment lengths are listed in Table I.

IV. METHODOLOGY

An overview of the Ukemi-SafeFall framework is shown in Fig. 2, in which we train a reinforcement learning policy via PPO to handle humanoid falling and recovery. The policy maps partial observations to target joint offsets and is trained

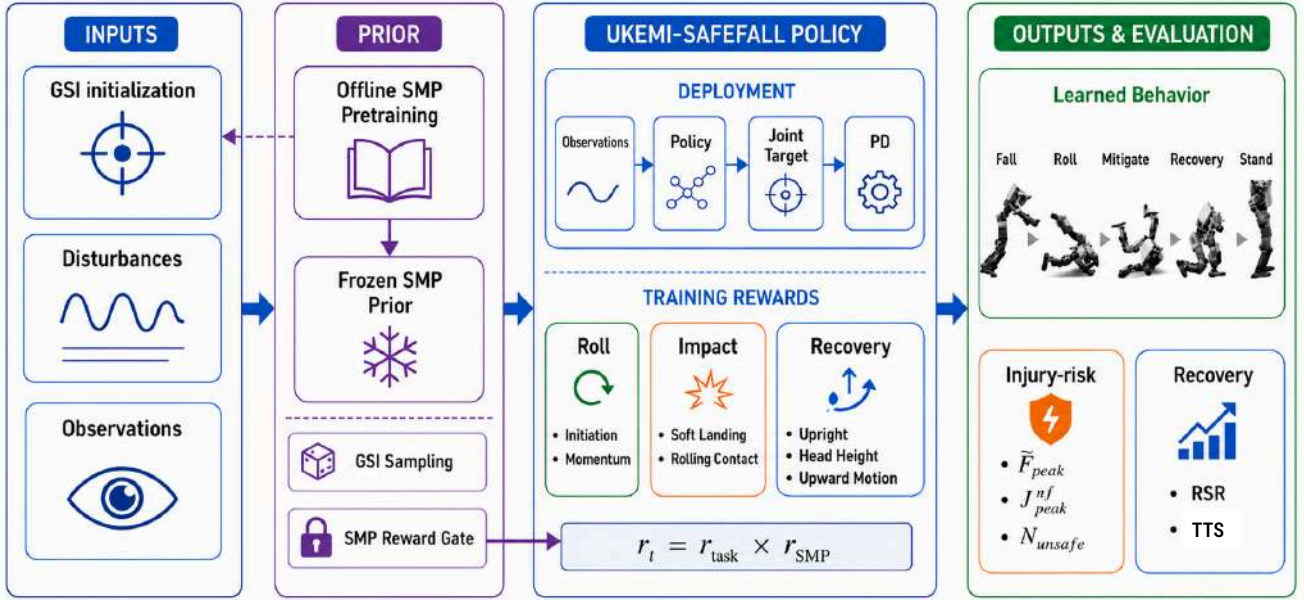


Fig. 2. Overview of Ukemi-SafeFall. A reinforcement learning framework trained via PPO to handle humanoid falling and recovery. The policy uses a multiplicative reward system that combines condition-gated task rewards (r_{task}), designed to encourage safe rolling and impact mitigation, with a diffusion-based human motion prior (r_{SMP})

TABLE I: Humanoid robot specifications

Parameter	Value	Unit
Max Torque: Hip, Knee, Ankle	120, 120, 36	Nm
Max Torque: Shoulder, Elbow	36, 36	Nm
Max Speed: Hip, Knee	13	rad/s
Max Speed: Shoulder, Elbow, Ankle	7.5	rad/s
Thigh inertia (I_{xx}, I_{yy}, I_{zz})	$(11.2, 10.6, 3.6) \times 10^{-3}$	kg m ²
Calf inertia (I_{xx}, I_{yy}, I_{zz})	$(9.9, 9.8, 2.0) \times 10^{-3}$	kg m ²
Ankle inertia (I_{xx}, I_{yy}, I_{zz})	$(2.1, 2.2, 0.59) \times 10^{-5}$	kg m ²
Upper-arm inertia (I_{xx}, I_{yy}, I_{zz})	$(0.4, 0.6, 0.3) \times 10^{-3}$	kg m ²
Lower-arm inertia (I_{xx}, I_{yy}, I_{zz})	$(0.3, 0.4, 0.2) \times 10^{-3}$	kg m ²
Body inertia (I_{xx}, I_{yy}, I_{zz})	$(7.3, 6.1, 3.2) \times 10^{-2}$	kg m ²
Hip to Knee, Knee to Ankle length	29.0, 26.5	cm
Shoulder to Elbow, Elbow to Wrist	19.6, 10.0	cm
Foot length, Hip to Shoulder length	20, 33.1	cm
Max Battery Voltage	50	V

with a multiplicative reward that combines condition-gated task rewards (encouraging safe rolling and impact mitigation) with a diffusion-based human motion-prior reward (r_{SMP}). To improve sim-to-real transfer and reduce injury-risk surrogates, the training pipeline exposes the agent to diverse fallen configurations and external disturbances through Generative State Initialization (GSI) and domain randomization.

A. Fall and Recovery Problem Formulation

We formulate humanoid falling and recovery as a finite-horizon Markov decision process (MDP) $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$ [30], where $\mathcal{S}$ and $\mathcal{A}$ denote the state and action spaces, $\mathcal{P}$ represents the system transition dynamics, $\mathcal{R}$ denotes the reward function, and γ is the discount factor. At each time step t , the agent observes a state s_t , selects an action $a_t \sim \pi(\cdot | s_t)$, receives a reward r_t , and transitions to a new state $s_{t+1} \sim \mathcal{P}(\cdot | s_t, a_t)$.

The objective is to learn an optimal policy π^* that manages non-foot ground contact, mitigates injury-risk surrogates, and

returns the robot toward an upright posture. In practice, the policy operates on an observation $\mathbf{o}_t$, which is a noisy projection of the true state s_t , i.e., $\pi(a_t | \mathbf{o}_t)$. For brevity, we omit the timestep t in the subsequent formulation.

Observation. We adopt an asymmetric actor-critic architecture: the actor receives only the partial observations available at deployment time, whereas the critic has access to additional privileged information during training. The actor's observation vector $\mathbf{o}$ is defined as

$$\mathbf{o} = [\omega^{\text{base}}, \mathbf{g}_{\text{proj}}, \boldsymbol{\theta} - \boldsymbol{\theta}^0, \dot{\boldsymbol{\theta}}, \mathbf{a}_{\text{last}}], \quad (1)$$

where ω^{base} denotes the base angular velocity, $\mathbf{g}_{\text{proj}}$ denotes projected gravity, $\boldsymbol{\theta} - \boldsymbol{\theta}^0$ and $\dot{\boldsymbol{\theta}}$ are the joint positions (relative to the default pose) and joint velocities, and $\mathbf{a}_{\text{last}}$ is the previous action. The critic's observation augments the actor's input with privileged variables:

$$\mathbf{o}^{\text{critic}} = [\mathbf{o}, \mathbf{v}_{\text{base}}], \quad (2)$$

where $\mathbf{v}_{\text{base}}$ denotes the base linear velocity.

Action. The policy outputs an action vector $\mathbf{a}$ as an offset from the default joint position $\boldsymbol{\theta}^0$. The target joint position is updated as

$$\boldsymbol{\theta}^{\text{target}} = \boldsymbol{\theta}^0 + \alpha \mathbf{a},$$

where α denotes a per-joint scaling factor. The target joint positions are then converted into joint torques using a proportional-derivative (PD) controller:

$$\boldsymbol{\tau} = \mathbf{K}_p (\boldsymbol{\theta}^{\text{target}} - \boldsymbol{\theta}) - \mathbf{K}_d \dot{\boldsymbol{\theta}},$$

where $\mathbf{K}_p$ and $\mathbf{K}_d$ are diagonal gain matrices.

Reward. The policy is trained to maximize a discounted return whose per-step reward is the product of two terms:

$$\mathbf{r} = \mathbf{r}_{\text{task}} \times \mathbf{r}_{\text{SMP}}.$$

Here, $\mathbf{r}_{\text{task}}$ is a weighted sum of ukemi-inspired task rewards, and $\mathbf{r}_{\text{SMP}}$ is a motion-prior reward computed from a pre-trained diffusion model using the SMP method [20]:

$$\mathbf{r}_{\text{SMP}} = \exp(-w_s \ell^{\text{SMP}}), \quad w_s = 6.0. \quad (3)$$

where ℓ^{SMP} is the mean denoising error over the selected diffusion steps (3 in our work). Lower error yields higher reward, indicating greater consistency with the prior motion distribution. The diffusion model is trained on ukemi fall-and-recovery motions from the CMU Graphics Lab Motion Capture Database [31], retargeted to the robot with GMR [32].

B. Ukemi-inspired Reward Design

Ukemi-SafeFall shapes fall and recovery behavior through condition-gated rewards based on the projected-gravity tilt ϕ and head height h_{head} , without using a discrete phase estimator. Formally, the task reward is defined as

$$\mathbf{r}_{\text{task}} = \sum_k w_k \mathbf{r}_k, \quad \sum_k w_k = 1,$$

where $\mathbf{r}_k$ denotes the condition-gated reward term defined in Table II, and w_k is its corresponding weight. Intuitively, when the robot becomes substantially tilted, the policy is encouraged to initiate rolling. Near the ground, additional terms encourage it to maintain rolling motion, reduce downward velocity, and limit peak contact force. Recovery terms encourage upright orientation, upward motion, and increased head height. Multiple reward gates may be active simultaneously.

TABLE II: Condition-gated reward terms

Reward term	Activation condition	Target	w_k
Recovery rewards			
Upright	Always	$\phi \rightarrow 0$	0.25
Head height	Always	$h_{\text{head}} \geq 1.08 \text{ m}$	0.20
Upward motion	$h_{\text{head}} < 0.83 \text{ m}$	$\dot{h}_{\text{head}} \geq 0.40 \text{ m/s}$	0.15
Ukemi rolling rewards			
Roll initiation	$\phi > 0.01 \text{ rad}$	$\ \omega_{xy}\ \geq 2.0 \text{ rad/s}$	0.20
Roll momentum	$h_{\text{head}} < 0.70 \text{ m}$	$\ \omega_{xy}\ \geq 1.5 \text{ rad/s}$	0.05
Impact-mitigation rewards			
Soft landing	$h_{\text{head}} < 0.50 \text{ m}$	Minimize $-v_z$	0.05
Rolling contact	$h_{\text{head}} < 0.70 \text{ m}$	$F_{\text{peak}} \leq 150 \text{ N}$	0.10

Here, $\|\omega_{xy}\|$ is the base angular velocity about the horizontal axes, v_z is the vertical base linear velocity, and F_{peak} is the largest link-level contact force at the current control step.

C. Injury-risk and Recovery Surrogate Metrics

To evaluate the proposed method, we introduce injury-risk surrogate metrics (normalized peak force, cumulative peak contact force, and unsafe-contact events) and recovery surrogate metrics (recovery success rate and time-to-stand). Fig. 3 illustrates these metrics before they are formally defined.

Let $\mathcal{C}$ denote the set of monitored links, and let $\mathcal{C}_{\text{nf}} \subset \mathcal{C}$ denote the subset of non-foot links. Let $\mathcal{C}_{\text{nf}}(t) \subseteq \mathcal{C}_{\text{nf}}$ denote

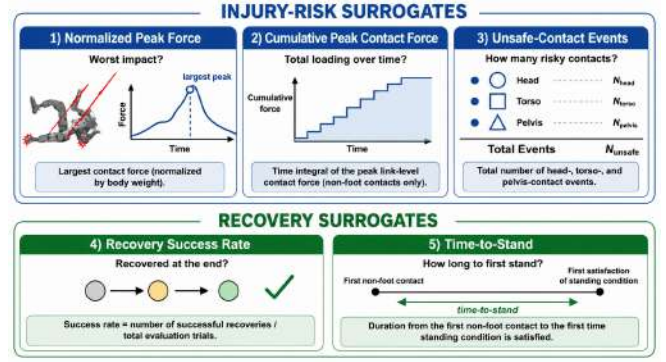


Fig. 3. Overview of the surrogate metrics used in this work.

the non-foot links in terrain contact at control step t . Let $\|\mathbf{F}_i(t)\|$ denote the contact-force magnitude measured on link i .

Normalized Peak Force. This metric is defined as follows:

$$\tilde{F}_{\text{peak}} = \frac{\max_t F_{\text{peak}}(t)}{9.81 \times m}, \quad (4)$$

where m is the robot mass and $F_{\text{peak}}(t) = \max_{i \in \mathcal{C}} \|\mathbf{F}_i(t)\|$. Intuitively, this metric is the largest contact force during an episode, expressed as a multiple of the robot's body weight. Lower values indicate less severe peak impact.

Cumulative Peak Contact Force. Unlike normalized peak force, cumulative peak contact force captures the accumulated peak link-level contact loading over time, reflecting both contact magnitude and duration. We define it as

$$J_{\text{peak}}^{\text{nf}} = \sum_{t: \mathcal{C}_{\text{nf}}(t) \neq \emptyset}^{T-1} F_{\text{peak}}(t) \Delta t_{\text{ctrl}}, \quad (5)$$

where $\Delta t_{\text{ctrl}} = 0.02$ and $F_{\text{peak}}(t)$ is computed over non-foot contact steps to avoid domination by foot contacts.

Unsafe-contact Events. Unsafe-contact events represent the total number of potentially hazardous head, torso, and pelvis contact occurrences during an episode. We define

$$N_{\text{unsafe}} = N_{\text{head}}^{\text{proxy}} + N_{\text{torso}} + N_{\text{pelvis}}, \quad (6)$$

Here, N_{torso} and N_{pelvis} denote raw contact onsets, recorded without force thresholds or temporal debouncing.

Head-contact Proxy Limitation. Because the simulated robot model lacks a rigid head collision body, direct measurement of head impacts is impossible. We use $N_{\text{head}}^{\text{proxy}}$ as a conservative *near-ground head* indicator, i.e., a surrogate. Specifically, $N_{\text{head}}^{\text{proxy}}$ is approximated by counting each instance in which the head drops below 0.50 m while any non-foot link touches the ground:

$$b_{\text{head}}(t) = \mathbf{1}\{h_{\text{head}}(t) < 0.50 \text{ m} \wedge \mathcal{C}_{\text{nf}}(t) \neq \emptyset\}. \quad (7)$$

Recovery Success Rate (RSR). An episode is counted as a recovery success when, at the final evaluation step,

$$h_{\text{head}} \geq 1.03 \text{ m}, \quad \|\mathbf{v}_{\text{base}}\| < 0.5 \text{ m/s}, \quad (8)$$

TABLE III: Domain randomization and observation noise.

Parameter	Range	Mode
External pushes		
Torso push force (+0.25 m)	± 200 N/axis	step
Torso push torque	± 20 N m	step
Push duration; cooldown	0.05–1.0; 1.5–4.0 s	step
Base linear-velocity push	$x, y: \pm 0.5; z: \pm 0.4$ m/s	interval
Base angular-velocity push	roll/pitch ± 0.52 ; yaw ± 0.78 rad/s	interval
Push interval	1.0–3.0 s	interval
Body and actuator dynamics		
Foot–ground friction	[0.3, 1.6] (shared)	startup
Torso CoM offset	($\pm 0.04, \pm 0.05, \pm 0.05$) m	startup
PD stiffness K_p (scale)	$\times [0.8, 1.2]$	reset
PD damping K_d (scale)	$\times [0.8, 1.2]$	reset
Joint encoder bias	± 0.015 rad	startup
Default joint offset	± 0.01 rad	startup
Observation noise		
Base angular velocity	± 0.2 rad/s	step
Projected gravity	± 0.05	step
Joint position	± 0.01 rad	step
Joint velocity	± 1.5 rad/s	step

where $\|\mathbf{v}_{\text{base}}\|$ denotes the magnitude of the linear velocity of the body.

Time-to-stand (TTS). Time-to-stand is the time from the first non-foot ground contact to the first satisfaction of Eq. (8), and is defined as

$$\text{TTS} = (t_{\text{stand}} - t_{\text{contact}}) \Delta t_{\text{ctrl}}. \quad (9)$$

Here, t_{contact} is the first control step at which any non-foot link contacts the ground. Both RSR and TTS are treated as recovery surrogates because the standing conditions only need to be satisfied once and are not required to remain satisfied afterward.

D. Domain Randomization and State Initialization

To enhance robustness and expose the policy to a wide variety of fallen configurations, we combine domain randomization with Generative State Initialization (GSI). We randomize the torso center of mass, foot–ground friction, PD gains (stiffness and damping), joint encoder biases, and default joint offsets. We also apply random pushes to the base velocity and random force–torque impulses to the torso. In addition, we inject noise to input observation. Domain randomization settings are provided in Table III. For GSI, the frozen SMP prior generates a pool of 4096 short motion windows; at each reset, one segment is sampled uniformly at random, and its final frame is used to set the base height, pitch/roll, joint positions, and joint velocities. The pool is partially renewed with 1024 new segments every 2400 environment steps. GSI is applied only during training.

V. EXPERIMENTS

A. Simulation and Experimental Setup

We built and evaluated Ukemi-SafeFall (USF) using Mjlab [33], a GPU-accelerated framework based on the MuJoCo physics engine [34]. All training and evaluation were

performed on a single NVIDIA GeForce RTX 5090 GPU across 4,096 parallel environments, with training completing in 20,000 iterations (approximately 7 hours). We trained the controller with PPO [35] using MLP policy (actor) and value (critic) networks with three hidden layers of 512, 256, and 128 units. The deployment code was implemented in C++; the policy was executed at 50 Hz using ONNX Runtime on the CPU. All hardware experiments were conducted on the humanoid robot described in Table I.

B. Fallen-State Evaluation Protocol and Metrics

All methods are assessed over $N=50$ simulated episodes using deterministic control actions, a fixed random seed, a control frequency of 50 Hz, and a time horizon of 5 s. Each episode starts from a specified fallen configuration, and the same set of initial configurations and random disturbances is applied to every method.

For the ablations, we report normalized peak contact force ($\tilde{F}_{\text{peak}}$), non-foot cumulative peak contact force ($J_{\text{peak}}^{\text{nf}}$), recovery success rate (RSR), and time-to-stand (TTS). For the cross-protocol reference comparison, we report $\tilde{F}_{\text{peak}}$, $J_{\text{peak}}^{\text{nf}}$, and unsafe-contact events (N_{unsafe}). Means and standard deviations are computed across evaluation episodes using one trained policy per method.

TABLE IV: Ablation and comparison methods.

Method	Ukemi r_{task}	Impact-mitigation	r_{SMP}
PF	✗	✗	✗
USF-R	✗	✗	✓
USF-NC	Always active	✓	✓
USF-NS	✓	✓	✗
SFT (*)	✗	✓	✗
USF (Ours)	✓	✓	✓

(*) Evaluated and reported as a cross-protocol reference

C. Comparison Methods

Using the same fallen-state evaluation protocol, we evaluate Ukemi-SafeFall under three ablations: Ukemi-SafeFall: Recovery Only, Ukemi-SafeFall without condition gating, and Ukemi-SafeFall without SMP reward gating. We also compare Ukemi-SafeFall against two baselines: Passive Fall and SafeFall Triangle PPO. Table IV summarizes the differences among the methods.

Passive Fall (PF). The controller outputs zero action after GSI initialization and is subjected to the evaluation disturbances. This baseline characterizes uncontrolled falling.

Ukemi-SafeFall: Recovery Only (USF-R). This ablation uses the same multiplicative SMP reward gate and GSI procedure as USF, but it is trained only with the upward-motion and head-height recovery rewards.

Ukemi-SafeFall without Condition Gating (USF-NC). This ablation retains the proposed ukemi-inspired rewards, contact-force shaping, and SMP gate, but disables the activation conditions; the corresponding reward terms remain active throughout the episode.

Ukemi-SafeFall without SMP Reward Gating (USF-NS). This ablation retains the condition-gated task rewards and contact-force shaping, but removes the multiplicative SMP reward gate. GSI continues to sample initial states from the SMP prior.

SafeFall Triangle PPO (SFT). We re-implement a SafeFall method from [3] to show how a protective falling policy performs without an explicit recovery objective. It is adapted to the same humanoid but uses a different training protocol; therefore, we report it as a cross-protocol reference rather than a baseline.

Ukemi-SafeFall (USF, Ours). The complete method combines condition-gated ukemi-inspired rewards and multiplicative SMP reward gating.

D. Simulation Results and Discussion

TABLE V: Comparison of injury-risk and recovery surrogate metrics among ablations.

Method	$\tilde{F}_{\text{peak}}$	$J_{\text{peak}}^{\text{nf}}$ (Ns)	RSR (%)	TTS (s)
USF-R	8.91 $\pm$ 4.89	319 $\pm$ 52	98.0	1.55 $\pm$ 0.48
USF-NC	12.44 $\pm$ 6.97	781 $\pm$ 268	36.0	3.95 $\pm$ 1.56
USF-NS	12.76 $\pm$ 5.55	1,411 $\pm$ 207	2.0	–
USF (Ours)	7.35 $\pm$ 5.11	179 $\pm$ 171	98.0	1.04 $\pm$ 0.27

Quantitative analysis. Under the same fallen-state evaluation protocol, Ukemi-SafeFall and USF-R both achieve a 98% recovery success rate (Table V), in contrast to 36% for USF-NC and 2% for USF-NS. For the successful episodes, Ukemi-SafeFall achieves the shortest mean TTS of 1.04 s, compared with 1.55 s for USF-R and 3.95 s for USF-NC. TTS is not reported for USF-NS because only one episode is successful.

Compared with USF-R, Ukemi-SafeFall maintains the same recovery rate while reducing mean TTS by 32.9%, mean $\tilde{F}_{\text{peak}}$ by 17.5%, and mean $J_{\text{peak}}^{\text{nf}}$ by 43.9%. Relative to USF-NC, Ukemi-SafeFall reduces the two contact-loading metrics by 40.9% and 77.1%, respectively, while increasing recovery success from 36% to 98%. Relative to USF-NS, the corresponding reductions are 42.4% and 87.3%, with recovery success increasing from 2% to 98%.

TABLE VI: Comparison on injury-risk surrogate metrics.

Method	$\tilde{F}_{\text{peak}}$	$J_{\text{peak}}^{\text{nf}}$ (Ns)	N_{unsafe}
PF	10.83 $\pm$ 6.09	828 $\pm$ 171	12.5 $\pm$ 7.1
SFT	7.24 $\pm$ 3.62	711 $\pm$ 50	2.9 $\pm$ 0.5
USF (Ours)	7.35 $\pm$ 5.11	179 $\pm$ 171	3.8 $\pm$ 3.6

Compared with PF, Ukemi-SafeFall reduces mean $\tilde{F}_{\text{peak}}$ by 32.1%, mean $J_{\text{peak}}^{\text{nf}}$ by 78.4%, and mean N_{unsafe} by 69.6% (Table VI). Fig. 4 complements these aggregate results by showing the episode-level peak-force distributions, including their medians, 90th percentiles, and empirical cumulative distributions.

Ukemi-SafeFall and SFT have similar mean normalized peak forces of 7.35 and 7.24, respectively, while Ukemi-SafeFall reduces mean $J_{\text{peak}}^{\text{nf}}$ by 74.8%. SFT records fewer

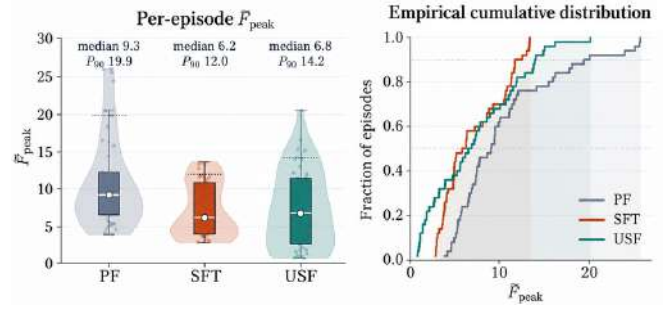


Fig. 4. Distribution of normalized peak contact force $\tilde{F}_{\text{peak}}$ for Passive Fall (PF), SafeFall Triangle PPO (SFT), and Ukemi-SafeFall (USF). Left: violin and box plots with the median (dot) and 90th percentile (dotted line). Right: empirical cumulative distribution functions. The median and 90th percentile are highlighted because the impact distributions are heavy-tailed.

unsafe-contact events on average (2.9 versus 3.8). Because SFT uses a different training protocol and does not optimize post-fall recovery, it is included only as a contextual safety reference.

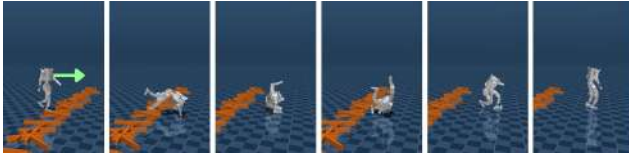
Discussion. The ablations indicate complementary contributions from the proposed components. USF-NC exhibits greater contact loading and a lower recovery rate, indicating the importance of activating fall-specific objectives only under their intended conditions. USF-NS performs poorly despite retaining the condition-gated task rewards, showing the importance of the SMP reward gate within the proposed training configuration. USF-R achieves frequent recovery, but the complete method further reduces contact loading and shortens recovery time.

Overall, Ukemi-SafeFall provides the strongest combined performance across the reported injury-risk and recovery surrogates. Because one trained policy is evaluated for each method, the reported standard deviations represent variation across evaluation episodes rather than uncertainty across independent training runs.

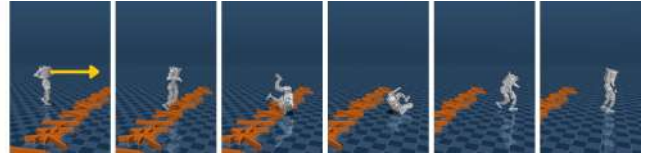
E. Sim-to-Sim and Sim-to-Real Results

We validate Ukemi-SafeFall across a range of falling scenarios. Fig. 5 illustrates several scenarios simulated in a separate MuJoCo-based environment, including falling while walking or running over obstacles, descending stairs, and jumping from a platform. Across these sim-to-sim validation scenarios, the controller exhibits rolling, distributed limb contact, and recovery toward standing.

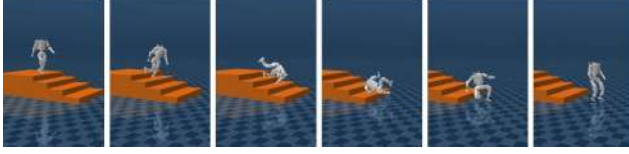
For hardware experiments, we additionally test the controller’s robustness by applying a kick disturbance while the humanoid is walking. Fig. 6 visualizes IMU orientation and torque utilization (% of rated torque limit) to provide a sim-to-real sanity check during execution of Ukemi-SafeFall. The time from fall onset to recovery is approximately 1 s. It can be seen that our method enables successful recovery without exceeding torque limits. Various hardware experiments are shown in Fig. 7, including a strong rear kick during walking,



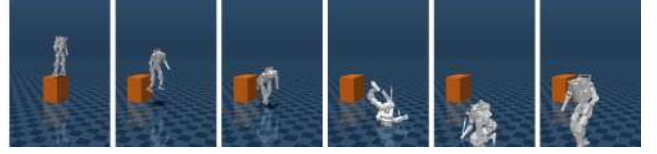
(a) Walking at 0.75 m/s over obstacles.



(b) Running at 1 m/s over obstacles.



(c) Walking at 0.75 m/s during going down the stairs.



(d) Jumping from a 0.6 m-high platform.

Fig. 5. Sim-to-sim results in several falling scenarios: (a) walking and (b) running over 0.08 m-high obstacles, (c) descending stairs, and (d) jumping from a 0.6 m-high platform.

walking over a 0.08 m strip obstacle, and jumping from a 0.6 m stair. These experiments show that the trained policy can be executed on the physical platform and qualitatively produces the intended fall–recovery sequence despite disturbances and sim-to-real mismatch.

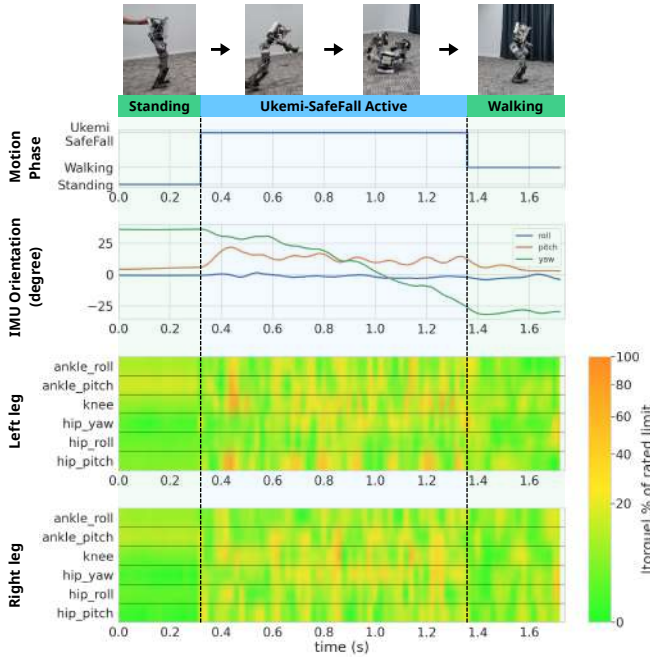


Fig. 6. **Hardware experiments.** Logged IMU orientation (roll, pitch, and yaw), and joint torque utilization (% of rated torque limit) during a physical fall and recovery.

VI. CONCLUSION AND FUTURE WORK

Conclusion. We present Ukemi-SafeFall, a ukemi-inspired, injury-aware falling and recovery framework. By combining condition-gated rolling and impact-mitigation rewards with an SMP-gated motion prior reward, the method manages the entire sequence from the initial fall to a post-fall recovery. Evaluated across diverse falling scenarios, Ukemi-

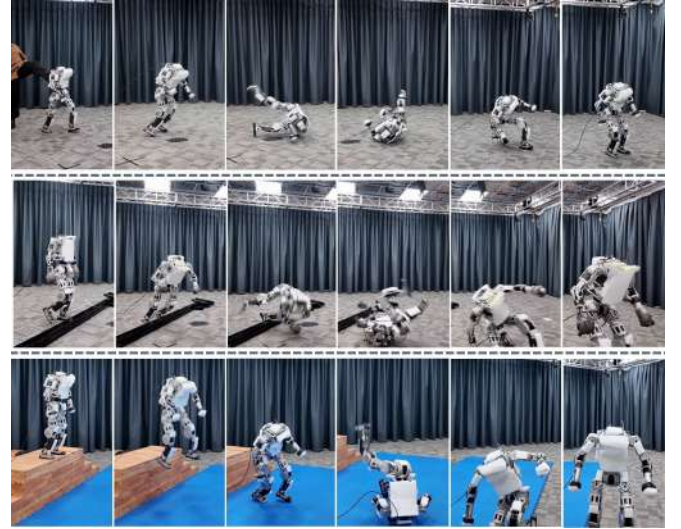


Fig. 7. **Hardware experiments.** From top to bottom: kicking while the humanoid is walking, walking/running over obstacles, and jumping from a platform.

SafeFall demonstrates promising results in both simulation, and real humanoid robot.

Future Work. Future work will focus on integration with the locomotion controller, potentially via hierarchical RL or an unified latent-space policy that can smoothly transition between walking gaits and falling behaviors. Crucially, our injury-risk metrics are structural surrogates computed in simulation. Because full-body rolling involves highly dynamic motions, hardware experiments are necessary to guarantee the protection of internal components. In addition, proxy measures such as $N_{\text{head}}^{\text{proxy}}$ capture only geometric proximity rather than true impact forces. Future work will deploy these policies on physical hardware to directly measure impact distributions and evaluate the framework on platforms with explicit head geometries.

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